A GROTHENDIECK SPECTRAL SEQUENCE FOR TWO SPECIALIZATION-STABLE SUPPORTS AND THE COMPOSITION OF THE ASSOCIATED DERIVED FUNCTORS

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Allowed standard prerequisites (editorial).

- The functor R\Gamma_T of sections with support in a specialization-stable subset T and its right adjointness to the inclusion of the category of modules supported in T. The author's section on more general supports, in which the counted statement occurs. Identifies the composite of the two support functors with the functor for the intersection of the supports.
- The Grothendieck spectral sequence of a composition of derived functors, in the form recorded in the chapter on derived categories. Invoked twice in the counted proof (statement line 1138-1139 and proof line 1158-1159) as a cross-chapter pointer. Produces the spectral sequence from the identification R\Gamma_{T \cap T'} = R\Gamma_T \circ R\Gamma_{T'}.

Source reference. Original reference derived-lemma-grothendieck-spectral-sequence: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1139–1139. Navigation pointer only: the counted text refers to the Stacks item derived-lemma-grothendieck-spectral-sequence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a ring and let $T, T' \subset \mathrm{Spec}(A)$ subsets stable under specialization. For $K \in D^+(A)$ there is a spectral sequence*

$$E_2^{p,q} = H^p_T(H^p_{T'}(K)) \Rightarrow H^{p+q}_{T \cap T'}(K)$$

as in Derived Categories, Lemma derived-lemma-grothendieck-spectral-sequence.

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Proof. Let E be an object of $D_{T\cap T'}(A)$. Then we have

$$\mathrm{Hom}(E, R\Gamma_T(R\Gamma_{T'}(K))) = \mathrm{Hom}(E, R\Gamma_{T'}(K)) = \mathrm{Hom}(E, K)$$

The first equality by the adjointness property of $R\Gamma_T$ and the second by the adjointness property of $R\Gamma_{T'}$. On the other hand, if $J^\bullet$ is a bounded below complex of injectives representing K, then $H^0_{T'}(J^\bullet)$ is a complex of injective A-modules representing $R\Gamma_{T'}(K)$ and hence $H^0_T(H^0_{T'}(J^\bullet))$ is a complex representing $R\Gamma_T(R\Gamma_{T'}(K))$. Thus $R\Gamma_T(R\Gamma_{T'}(K))$ is an object of $D^+_{T\cap T'}(A)$. Combining these two facts we find that $R\Gamma_{T\cap T'} = R\Gamma_T \circ R\Gamma_{T'}$. This produces the spectral sequence by the lemma referenced in the statement. $\square$